# Deep Learning

## Lecture 4: Adversarial models

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# Lecture overview

## 1 Generative adversarial networks

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- properties
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- Lipschitz continuity
- spectral normalisation
- conditional GANs
- information maximizing GANs
- modern GAN training
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- super resolution
- adversarial anomaly detection

## 3 Attacks and detection

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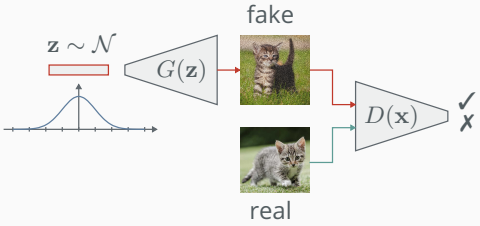
# Generative adversarial networks definition

**Definition:** generative adversarial networks

A generative adversarial network (GAN) is a non-cooperative zero-sum game where two networks compete against each other [1].

One network $G(\mathbf{z})$ generates new samples, whereas $D$ estimates the probability the sample was from the training data rather than $G$:

$$\min_G \max_D V(D,G) = \mathbb{E}_{\mathbf{x}\sim p_{\text{data}}(\mathbf{x})}[\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z}\sim p_{\mathbf{z}}(\mathbf{z})}[\log(1 - D(G(\mathbf{z})))].$$

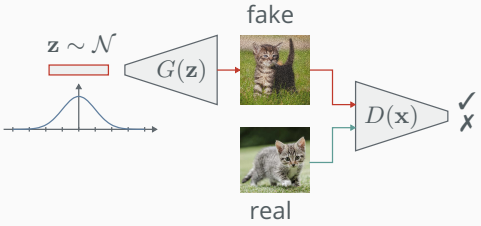


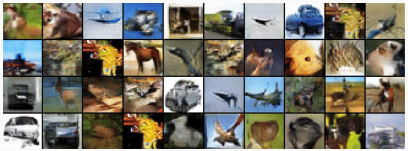
## GAN **properties**

GANs benefit from differentiable data augmentation [2] for both reals and fakes, but are otherwise notoriously difficult to train:

- Non-convergence
- Diminishing gradient
- Difficult to balance
- Mode collapse (next slide)

**Link to Colab example**



**Definition:** mode collapse

This is where the generator gets stuck producing, or rotates through, a small subset of outputs, and the discriminator is unable to get it out of the trap. Mode collapse is arguably the main limitation of GANs.

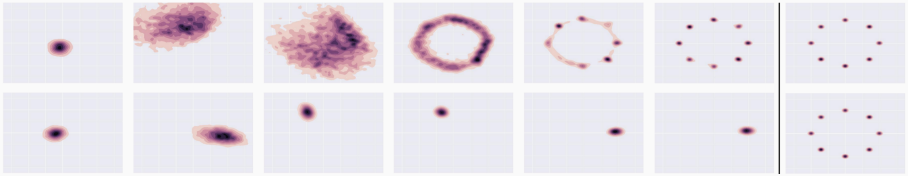


Figure from [3]. The final column shows the target data distribution and the bottom row shows a GAN rotating through the modes.

**Definition:** Lipschitz function

A function $f$ is Lipschitz continuous if it is bounded by how fast it can change. Specifically if there exists a positive real constant $k$ where:

$$|f(x) - f(y)| \leq k|x - y|,$$

for all $x, y$ in the domain. For example, any function with a bounded first derivative on an interval is Lipschitz on that interval.

Wasserstein GANs [4, 5] use a $1$-Lipschitz discriminator (called a critic) while optimising the Wasserstein objective.

Distance functions are 1-Lipschitz

## Definition: spectral normalisation

The matrix (spectral) norm defines how much a matrix can stretch a vector $\mathbf{x}$:

$$\|\boldsymbol{A}\| = \max_{\mathbf{x}\neq 0} \frac{\|\boldsymbol{A}\mathbf{x}\|}{\|\mathbf{x}\|}$$

Spectral normalisation [6] divides each weight matrix by its spectral norm $\sigma(\boldsymbol{W})$. For a sequence of linear layers with $1$-Lipschitz activations, this bounds the network's Lipschitz constant:

$$\hat{\boldsymbol{W}}_{\text{SN}}(\boldsymbol{W}) = \boldsymbol{W}/\sigma(\boldsymbol{W})$$
$$\sigma(\hat{\boldsymbol{W}}_{\text{SN}}(\boldsymbol{W})) = 1$$
$$\|f\|_{\text{Lip}} \leq 1$$

## Spectrally normalised discriminator

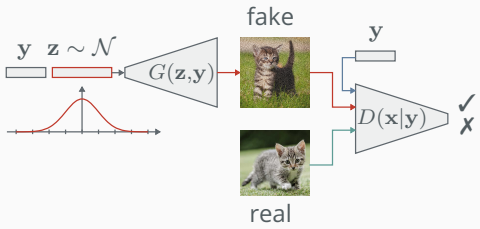
```
class Discriminator(nn.Module):
  def __init__(self, f=64):
    super().__init__()
    self.discriminate = nn.Sequential(
      spectral_norm(Conv2d(1, f, 3, 1, 1)),
      nn.LeakyReLU(0.1, inplace=True),
      nn.MaxPool2d(kernel_size=(2,2)),
      spectral_norm(Conv2d(f, f*2, 3, 1, 1)),
      nn.LeakyReLU(0.1, inplace=True),
      nn.MaxPool2d(kernel_size=(2,2)),
      spectral_norm(Conv2d(f*2, f*4, 3, 1, 1)),
      nn.LeakyReLU(0.1, inplace=True),
      nn.MaxPool2d(kernel_size=(2,2)),
      spectral_norm(Conv2d(f*4, f*8, 3, 1, 1)),
      nn.LeakyReLU(0.1, inplace=True),
      nn.MaxPool2d(kernel_size=(2,2)),
      spectral_norm(Conv2d(f*8, 1, 3, 1, 1)),
      nn.Sigmoid()
    )
```

**Definition:** conditional GAN

GANs can be conditioned with labels $\mathbf{y}$ if available [7] by feeding the label information into both the generator and the discriminator:

$$\min_G \max_D V(D,G) = \mathbb{E}_{\mathbf{x},\mathbf{y}\sim p_{\text{data}}(\mathbf{x})}[\log D(\mathbf{x}|\mathbf{y})] + \mathbb{E}_{\mathbf{z}\sim p_{\mathbf{z}}(\mathbf{z})}[\log(1 - D(G(\mathbf{z},\mathbf{y})|\mathbf{y}))].$$

**Link to Colab example**



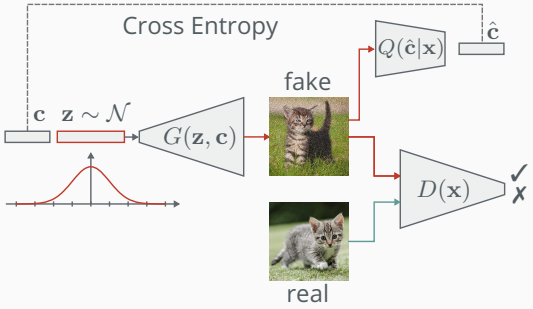
**Definition:** information maximizing GANs

GANs can be trained to learn disentangled latent representations in a completely unsupervised manner. InfoGAN [8] popularised this by maximizing mutual information between the observation and a subset of the latents:

$$\min_{G,Q} \max_{D} V_{\text{InfoGAN}}(D, G, Q) = V(D, G) - \lambda L_I(G, Q)$$

where $L_I(G, Q)$ is a variational lower bound of the mutual information.

**Link to Colab example**



**Definition:** relativistic GAN

Replace the usual adversarial loss with a comparison of raw discriminator scores:

$$\min_G \max_D \mathop{\mathbb{E}}_{\substack{\mathbf{z}\sim p_{\mathbf{z}} \\ \mathbf{x}\sim p_{\text{data}}}} \left[\log \sigma(D(G(\mathbf{z})) - D(\mathbf{x}))\right],$$

where $\sigma$ is the sigmoid.

R3GAN [9] adds penalties to $D$'s update for large score changes from tiny image changes. GAT [10] extends this with Transformers in a pretrained image latent space.

GAT-XL/2: curated ImageNet-256 samples. Hyun et al., ICML 2026 [10].

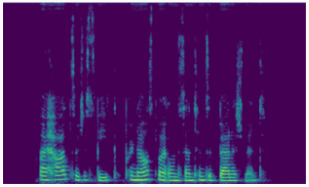
## "Stick a GAN on the end"

A pretrained generator is further trained with a discriminator to improve the realism of its outputs.

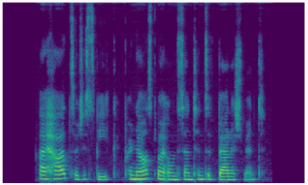
Flow2GAN [11] applies this to a flow-matching model that synthesises speech audio from a mel-spectrogram (a compact description of the sound).

**Listen to the examples**

Before adversarial fine-tuning



After adversarial fine-tuning



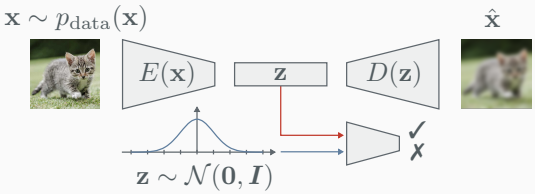
Generated audio (spectrograms, two steps).

Yao et al., ICLR 2026, Fig. 2(e–f) [11].

**Definition:** adversarial autoencoders

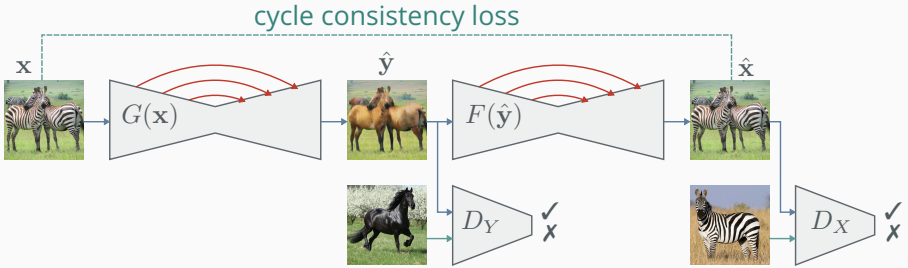
Adversarial autoencoders [12] are generative models that permit sampling.

In addition to the reconstruction loss, such $\|\mathbf{x} - \hat{\mathbf{x}}\|^2$, they use adversarial training to match the aggregated posterior of the hidden code vector $\mathbf{z}$ of the autoencoder with an arbitrary prior distribution, such as $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$.



**Definition:** unpaired translation (CycleGAN)

CycleGAN [13] propose an adversarial architecture that enables unpaired image translation. It has twin residual generators and two discriminators, which translate between the domains, alongside a cycle consistency loss (an L1 norm) which ensures the mapping can recover the original image.



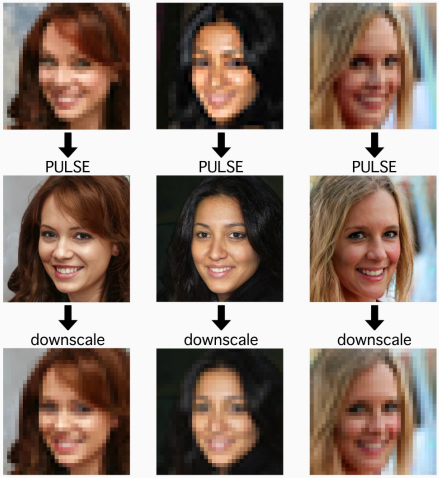
# Popular applications super-resolution

**Definition:** super-resolution

Adversarial models are popular in super-resolution approaches. The challenge is that a single low-resolution (LR) input can map to a distribution of high-resolution (HR) outputs.

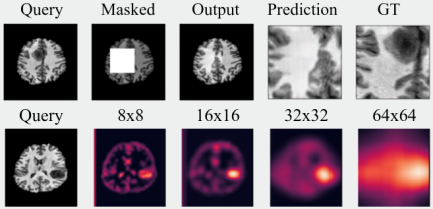
PULSE [14] searches the latent space of a pretrained StyleGAN for outputs matching the LR input after downsampling. It projects latent points onto a hypersphere to keep their radius typical of the Gaussian prior.

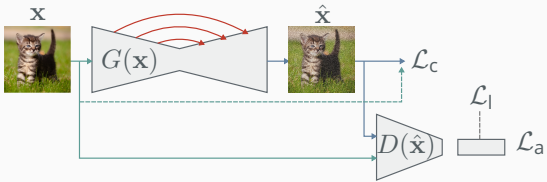
**Online example**



**Definition:** anomaly detection

Unsupervised anomaly detectors [15] learn a distribution over normal (healthy) observations. Then, when they observe something not observed in training (unhealthy/dangerous), they may reconstruct it poorly - detecting it as an anomaly. Region-based anomaly detectors [16] learn a distribution over inpainted (erased) regions.



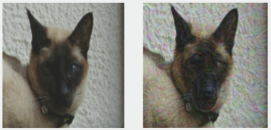



# Adversarial examples attacks

## **Definition:** adversarial examples

These are small but intentionally worst-case perturbations that fool the model to output incorrect answers with high confidence [17]. It is possible to generate examples that also fool the human visual system [18]. Cat or dog?

original adv



## **Example:** adversarial examples

Example adding an imperceptibly small vector by the sign of the elements of the gradient of the cost function with respect to the input [17]:

**Question:** who wins at the limit?

Deepfake generators improve, and detectors learn to spot their remaining artefacts.

If generated and real images have the same distribution [1],

$$p_G = p_{\text{data}} \quad \Longrightarrow \quad D^*(\mathbf{x}) = \tfrac{1}{2}.$$

An image-only detector is then at chance on balanced real/fake samples.

## Watermarking

Embed an intentional signal during generation, e.g. **SynthID**, so realistic images can still be detected.

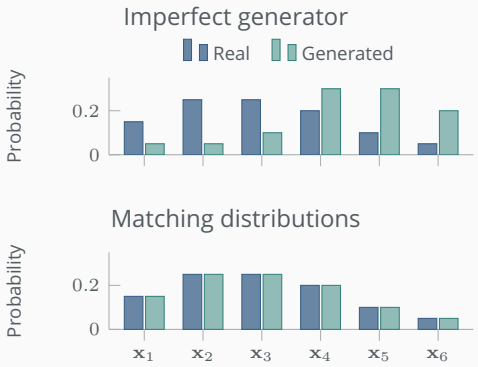



Illustration with six possible images.

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# References IV


[15] Samet Akçay, Amir Atapour-Abarghouei, and Toby P Breckon. **"Skip-ganomaly: Skip connected and adversarially trained encoder-decoder anomaly detection".** In: 2019 International Joint Conference on Neural Networks (IJCNN). IEEE. 2019, pp. 1–8.

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